



Abrar Fahim

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About myself

Electrical Engineering graduate with a strong foundation in power systems, energy modeling, and optimization. Experienced in deep learning, power systems optimization and several machine learning projects using MATLAB and Python. Current research interests include AC optimal power flow, and smart grid operation along with computer vision. Motivated to pursue advanced studies in power engineering with modern machine learning architectures.

Education

B.Sc. in Electrical & Electronic Engineering | CGPA: 3.44/4.00

Islamic University of Technology | 2 Jan 2020 - 7 May 2024

HSC (Science) | GPA 5.00/5.00 | Chittagong College

SSC (Science) | GPA 5.00/5.00 | Government Muslim High School

Under Review & In Progress Papers

Comparative Analysis of Classical , Machine Learning, Deep Learning, and Metaheuristic Approaches for Optimal Power Flow | 2nd IEEE ICEESI 2026, India

- Conducted a unified comparative study of AC optimal power flow solved using classical OPF, Random Forest, DNN, and Particle Swarm Optimization on IEEE test systems

Design and Implementation of a Solar Energy-Powered Smart Street Lighting Control System for Sustainable Urban Development in Bangladesh | IEEE IPRECON 2026, Kerala, India

- Designed a solar-powered smart street lighting control system integrating automated energy management and control to reduce power consumption and support sustainable infrastructure in Bangladesh.

Enhancing Intrusion Detection in SDNs Using Hybrid Feature Selection and XGBoost | Pending Submission

- Trained a hybrid intrusion detection system using XGBoost and hybrid feature selection (CFS + RF-RFE) for Software-Defined Networking (SDN), achieving 99.97% accuracy on the NSL-KDD Dataset.

Transparent and Accurate Anemia Prediction Through GA-Selected Features and Grid-Search Enhanced Decision Trees | Pending Submission

- Proposed an interpretable anemia prediction model using Genetic Algorithm–based feature selection, achieving high predictive accuracy while preserving clinical transparency and explainability.

Conference Proceedings

Deep Learning Based Framework for Colorectal Cancer Using Histopathological Images | 7th ICAIIC 2025, JAPAN

- Developed a framework using advanced pre-processing techniques for detecting colon cancer.
- Used a modified MobileNetV2 for classification achieving 99.95% accuracy on the LC25000 dataset.

Optimized Approaches to Malware Detection: A Study of Machine Learning and Deep Learning Techniques 14th IEEE CSNT 2025, VIT Bhopal, India.

- Trained ML models like Multi-Layer Perceptron, KNN, Logistic Regression & SVM for malware detection
- Achieved 99.97% accuracy by optimizing the preprocessing and used algorithms like CNN, DNN, and Random Forest

A Modified VGG19-Based Framework for Accurate and Interpretable Real-Time Bone Fracture Detection
16th ICCCNT 2025, IIT-Indore, Madhya Pradesh, India.

- Developed a modified VGG19-based deep learning framework for automated bone fracture detection in X-ray images, achieving exceptional 99,78% classification accuracy.
- Applied CLAHE, Otsu's thresholding and Canny edge detection, outperforming comparison models

Analysis of Pre-trained CNN Models in MRI-Based Brain Tumor Detection
27th IEEE ICCIT 2024, Cox's Bazar.

- Performed a comparative analysis of five CNN models (InceptionV3, ResNet-50, VGG-16, MobileNerV2, DenseNet121) for classifying brain tumors in MRI images and performed both binary and multi-class classification

Predicting Antibiotic Resistance in Gonorrhea: A Comparative Analysis of Machine Learning, Deep Learning, and Explainable AI |
27th IEEE ICCIT 2024, Cox's Bazar, Bangladesh.

- Applied XAI (GRU, LSTM, RNNs) for antibiotic resistance prediction.
- Applied deep learning techniques like CNN, Random Forest and Decision Tree for Neisseria gonorrhoeae, showing high prediction accuracy and potential solutions for antibiotic resistance in public health.

Diabetic Retinopathy Diagnosis Using a Hybrid Efficient Net-ResNet Model with Coordinate Attention
IEMTRONICS 2025, Springer Nature, Imperial College London, UK

- Developed hybrid EfficientNet-B3 ResNet-50 model with Coordinate Attention
- Achieved 83.61% accuracy, 93.34% accuracy and 0.9051 Kappa Score on the APTOS 2019 dataset

An Optimized Support Vector-Based Hybrid Framework for Real-Time Maternal Risk Prediction: GA-Driven Feature Selection and XAI
16th ICCCNT 2025, IIT-Indore, Madhya Pradesh, India.

- Applied GA-driven feature selection with SVC for maternal risk prediction (98.21% accuracy, 0.9903 AUC)
- Used LIME to identify key risk factors and developed web app for real-time risk stratification

VisionEdge: Cataract Detection via Smartphone
27th IEEE ICCIT 2024, Cox's Bazar, Bangladesh.

- Developed VisionEdge, a mobile app using MobileNetV2 for real-time cataract detection with 99.11% accuracy, optimized for Edge Intelligence, providing a cost-effective and accessible solution for cataract screening.

Language Skills

Mother tongue(s): **Bengali**

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	C2	C2	C1	C1	C1

Work experience

Risk & Operational Analyst | NEXT Ventures | Department of Trading & Risk Management | 16/04/2025 - Current | Dhaka, Bangladesh

- Currently promoted to Trading Platform Engineer & handling entire trading backend + server + software + execution feed & latency.
- Responsible for the launch of the newest trading brokerage FNmarkets and their CRM platform.
- Ensure uninterrupted trading operations by managing platform settings and optimizing trade execution throughout the entire system.
- Oversee risk control measures, including real-time market monitoring and trade routing to liquidity providers.
- Develop and deliver comprehensive analytics, dashboards, and reports to support strategic decision-making.
- Collaborate closely with internal teams including Support, Product, Back Office, and Tech-to resolve trade disputes, and maintain seamless operations

Data scientist | Synnax Laboratory, Credit Intelligence Division | 01/05/2025 - Current | Dhaka, Bangladesh

- Developed end-to-end machine learning SDK & pipelines for multi-target financial forecasting, including robust data preprocessing, feature engineering, and ensemble of multiple models.
- Regular data prediction, hyperparameter tuning, and leaderboard submission workflows, achieving top-quartile accuracy on proprietary financial datasets.

Project Engineer (Electrical) | Mark Enterprise, BPDB Contractor | 08/06/2024 - 09/12/2024 | CHATTOGRAM, Bangladesh

- Overseeing procurement related agreement papers from several stakeholders & vendors.
- Creating comparative statements in Excel and databases along with data visualization.
- Collaborating with civil engineers to handle the project plans in different locations.

Skills

Presentation | Documentation | Event Management | Teamwork | Critical Thinking

Data Analysis | Machine Learning (AI, DL, CNN) | Python (NumPy, Pandas, NLTK, Matplotlib, sklearn, TensorFlow) | Data Science | Data Collection, Data Processing, Data Analysis, Data Visualisation | SQL | Frameworks and Libraries: FastAPI, Open CV, Tensorflow, Keras, Pandas, Seaborn, Plotly, Streamlit)

Languages | LaTeX | MATLAB | Assembly (computer programming) | CSS & HTML | TypeScript

Software | Google Workspace | Visual C - Visual Studio | Microsoft Office | Management Software - Clickup | TinkerCAD | Mendeley Citation Management | Notion

Certifications

Course Certificates

- Introduction to Data Science - [Simplilearn]
- Introduction to Neural Network - [Simplilearn]
- Google AI Essentials - [Coursera]
- Introduction to Python OpenCV - [Simplilearn]
- Machine Learning using Python - [LinkedIn Learning]
- What Is Generative AI? - [Simplilearn]
- Image Recognition Basics for Beginners - [Simplilearn]

Projects

Plasmonic Coupling & Thermal Effects on Photothermal Response of Randomly Distributed Nanoparticles
Presented at the International Conference on Centennial Celebration of Bose-Einstein Statistics

- Investigated the influence of plasmonic coupling on the photothermal behavior of silver nanoparticles.
- Used MATLAB and Python for illustrating temperature in nanoscale, and view absorption spectra.

Economic Load Dispatch Problem Using Python & MATLAB

Python, MATLAB, LaTeX

- Performed Economic Load Dispatch (ELD) Calculation for a Bangladeshi Power Grid Data using MATLAB
- Predicted the accuracy of several numerical methods using machine learning models via python.

SecureQRGen: Double Layer Encrypted QR Code Generator

Python, PyQt5

- Developed a Python-based QR code generator using base64 encryption and decryption.
- Used PyQt5 for building a user friendly GUI for functions like scanning and transmitting.

AI Sudoku Solver Using Constraint Satisfaction |

Github, Python, PyQt5

- Developed a Sudoku solver incorporating a PyQt5-based interface for real-time interaction and puzzle-solving.